Part II of the Scientific Proposal

1. Implementation overview

Part I introduces the five work packages and their order. Here we describe how they are coordinated. The order is not a waterfall: WP1 delivers its first account of the required islands at M12 and revises it annually; WP5 involves each island's community from M12 onward; and WP2 does not hand down a finished specification, since we revise the calculus whenever WP3 and WP4 report back.

Integration. Each project year closes with an integration milestone at which we build the WP3 and WP4 prototypes into a single running *island* stack and deploy it at one or more WP5 sites. Each milestone produces a versioned reference implementation, a revised WP1 account and decision record, and a written *calculus reconciliation*, which records for every architectural decision taken that year whether it was derivable from the current WP2 model or forced a revision of it. The reconciliation makes the theory falsifiable once a year, on a schedule fixed in advance.

Layering. WP1 fixes the NFP bundles; WP2 names them as link classes; WP3 defines how each class is upgraded, downgraded, or refused at a border; WP4 supplies the addressing and *transclusion* substrate (T4.1), turns SMPC-based resolution into an addressing primitive that the link-class typing can name (T4.2), and exposes both to agent clients as state-dependent *affordances*, or binds them at design time as *Static Links* (T4.3). WP5 instantiates the whole in islands whose threat models, stakes, and governance differ; this difference is what allows the WP2 model to claim generality.

Openness. We release all software as open source under a permissive licence, submit all specifications to the W3C groups that maintain the protocols islands build on, or to their successors (Linked Web Storage WG, Social Web WG and CG, Verifiable Credentials and Federated Identity WGs, Web of Things WG, Autonomous Agents on the Web CG; RFC 9518 advises reusing such proven mechanisms), publish open access, and publish all benchmark harnesses and datasets that are not legally restricted alongside the papers.

2. Methodology

2.1 WP1 — Societally required islands (M1–M18, revised annually)

Method: stakeholder-based requirements analysis with design-science iteration.

Island profiles. WP1 starts from the five island drafts of Part I (Figure 1) and turns each into an *island profile*: the community and its stakes, the NFP bundle, the actor that keeps each guarantee, the threat model, and the governance of admission. Two drafts raise questions of their own. For Island F, WP1 checks that binary bindings can stay compatible with the Web architecture, i.e., that its resources keep URIs, are described in WoT Thing Descriptions, and are reachable through the uniform interface at the border. For Island N, whose nanopublication network already provides immutable, signed, and replicated claims (Kuhn and Dumontier 2014; Kuhn et al. 2016; Nanopublication Association 2026), WP1 asks how an existing network can become an island without migrating its content, which tests the claim that islands extend the infrastructure that communities already run.

Stakeholder analysis. We study each class of actor (hyperscaler, legacy industry, OSS/privacy community, browser vendor, platform, and public record-keepers such as parliaments, courts, and national archives) through semi-structured interviews **[TODO: target n per class; ethics approval route at USG]**, institutional analysis of funding models, the RFC 9518 mitigation checklist, and the W3C TAG's societal-impact self-review questionnaire (Capadisli et al. 2026), read against the European Commission's DSA enforcement record, including its December 2025 decision against X on deceptive verification badges and researcher data access. For platforms, the analysis also covers licensing of content to AI model providers as a new incentive to close access, as in Reddit's move to a primarily logged-in experience after AI answers cut its search referrals (Reddit 2026). For each island and each class, we ask which NFP bundle the community needs, which actor would keep each guarantee, whether the island changes that actor's incentive to centralize, and whether the island can be configured so that the guarantee-respecting path is the actor's least costly one; the last question supplies WP2's economic cost. The analysis builds on our published platform-transparency and DSA data-access work, on the PI's seat on the European VLOP-auditing oversight board, and on our working relationship with the ODI through CoCoDa. For the legacy-industry class, which is otherwise the hardest to recruit, the PI's role in a 33-partner Chips JU consortium of semiconductor, edge-platform, and system-integration vendors provides named interviewees (§3).

Business, legal, and economic analysis. Each island draft is also analysed as an institution, not only as an architecture. The business analysis asks who operates an island, who bears its coordination cost, and which decentralized business model (e.g., membership fees, cost sharing among members, or public funding for public-record islands) can sustain it; it draws on Tim Meyer's work on decentralized business models at HSG.

The legal analysis asks which obligations an island's NFPs must meet and which they help meet: copyright and neighbouring rights for Island L, including the quotation right (Directive 2001/29/EC, Art. 5(3)(d)) and the reservation against text and data mining (Directive (EU) 2019/790, Art. 4(3)) that providers of general-purpose AI models must respect under the AI Act (Regulation (EU) 2024/1689, Art. 53(1)(c)), the battery passport and CBAM for Island S, the DSA for Island M, GDPR Art. 17 for every island that keeps immutable records, and the liability of island operators; it runs through the Future Society Hub (FuSo), where our group has long worked with legal scholars, and with Prof. Tamò-Larrieux. The economic analysis asks how an island changes costs, network effects, and switching costs, and it supplies the compliance margins that enter WP2's economic cost **[TODO: name the economics collaborator at HSG]**. The three analyses feed the account of required islands and the decision record: an island without a sustainable business model, or whose guarantees conflict with the law of its domain, is recorded as an exclusion.

Architectural decision record. WP1 records what ISLANDS does not do, in the established format of architectural decision records (context, decision, consequences). Each exclusion states the island it rules out and the reason: islands that need Web-wide coordination (e.g., a global registry that replaces DNS); islands without a downgrade path to the open Web; islands that enforce usage on devices their members do not control, i.e., islands that would act as digital rights management by restricting reading or copying rather than making reuse traceable; and islands whose attribution regime the stakeholder analysis finds incompatible with fundamental rights. The record bounds the theory as much as the account of required islands does, since every exclusion is a claim that the calculus does not need to cover.

Deliverables. The account of societally required islands, each an NFP bundle with its community, stakes, threat model, and the actor that keeps each guarantee (M12, revised annually); the architectural decision record (M12, revised annually); and the stakeholder dataset, published where consent allows.

2.2 WP2 — Link classes and the guarantee calculus (M6–M60)

Method: formal modeling with design-science iteration.

Construction. WP2 isolates the *link classes* that the WP1 *islands* require and orders them in a lattice of guarantee bundles. We annotate each bundle with (i) its minimal coordination requirements, expressed as standing obligations on island members (registration, replication factor, identity binding, availability coupling); (ii) the threat models under which it is admissible; and (iii) its cost vector. Information-flow control approached the problem from programming languages. The decentralized label model (Myers and Liskov 1997) and Fabric (Liu et al. 2009) let mutually distrusting principals share persistent objects under confidentiality and integrity labels. They enforce these labels by placing cleartext only on trusted nodes, over mutable, store-identified objects, and they do not quantify what sustaining a policy costs. Our formalization follows this tradition: a *link class* is a label on a lattice in the style of the decentralized label model, which we extend from confidentiality and integrity components to persistence and attribution components, so that ordering, join, and the proof techniques of label lattices carry over. We express persistence and integrity components as safety and liveness obligations over a content-addressed store; confidentiality components reuse the simulation-based definitions of T4.2.

Link classes. Table 1 fixes the link-class vocabulary that WP3 translates and T4.1's structural typing enforces.

Class	Axis	Resolution guarantee	Minimal coordination; precedent
Classic Jumplink	ladder	nothing: location-addressed, mutable	none; the Web default and the fallback outside any island
Archived/Resilient Link	ladder	stays resolvable across edits and removals; not verified	redundant snapshots and signatures (LOCKSS, Memento, Perma.cc, robust links)
Trusted Link	ladder	registrant known and accountable; bidirectionally tracked	island link registry (Microcosm/DLS lineage)
Verified Link	ladder	the exact cited span at the pinned version, or visible failure	content-addressed store (Git/IPFS, SWHID)
Cryptographic Link	ladder	a verified derivation from data the client never sees	secret-shared targets, SMPC/ZK resolution (PEP-LCA)
Attributed Link	attribution	reference bound to a verified author	identity binding (PROV, C2PA, eIDAS 2.0 wallets)

Class	Axis	Resolution guarantee	Minimal coordination; precedent
Anonymous/Pseudonymous Link	attribution	eligibility verified without identity	anonymous credentials, revocable pseudonyms
Credentialed Link	trust	an issuer's *verifiable credential* about the source, with its current revocation status	W3C Verifiable Credentials 2.0, SD-JWT (RFC 9901), eIDAS 2.0 wallets; island-defined list of accepted issuers
Endorsed/Reputation Link	trust	decay-bounded, Sybil-resistant score returned with the target	retractable endorsement web (nanopublication IDEBT)
Purpose-Bound Link	usage	derivation valid for a stated recipient, purpose, and window; re-resolved on expiry	terms carried in the reference, in ODRL 2.2 with DPV purposes; assessed by compliance margin (transcopyright, insight economy)
Capability/Affordance Link	modality	an action, exposed only while valid; any rung applies	T4.3 agent gateway (*signifiers*)
Static Link	*link time*	interface bound at design time in a fixed binary encoding; no run-time discovery; fails visibly if either side changes	design-time agreement on schema and encoding; static linking (Levine 2000); WoT TD binary protocol bindings (e.g., Modbus), WoT Binding Templates
Live/Subscription Link	narrower	notification when the held version is superseded	WebSub-style hub per island
Personalized/Disclosed Link	narrower	resolution varies by requester, with the criteria disclosed	disclosed-filter design; counterpart to undisclosed platform curation

Table 1. The link-class family that WP2 formalizes, WP3 translates, and T4.1 enforces. Retraction is a state, not a class: content stays immutable in the substrate, and a superseding retraction record changes the default resolution. This is how a Verified or *Archived Link* honours GDPR Art. 17 without breaking the history on which other references depend.

Proof obligations. For each link class, the model must discharge *realizability* and *necessity*. *Realizability:* given an island that satisfies coordination requirement set C under threat model T, a conforming implementation that resolves a reference of class L either returns a result meeting L's guarantee or fails visibly, with no silent downgrade. *Necessity:* dropping any element of C admits an execution in which L's guarantee is violated without visible failure. Necessity keeps the ladder honest, since it states that a rung cannot be attained at lower coordination cost.

Coordination cost and economic cost. We measure coordination cost in the standing obligations above and instrument it directly in the T4.1 benchmarks. Economic cost enters with the *usage* axis: a Purpose-Bound Link's terms cannot be enforced after resolution, so the model scores such bundles by the margin by which compliance is less costly than defection for the party able to defect. Operationally, we compare the cost of re-requesting a fresh derivation with the cost of storing, indexing, and re-using a stale one, per actor class and from the WP1 stakeholder data. Transcopyright set the terms of reuse in this way, and current data-economy proposals set the terms of derived insights in the same way; our contribution is to treat the margin as a term in a *guarantee calculus*, so that the theory can return "this guarantee is not attainable for this actor" as a first-class answer.

Falsification protocol. At each annual integration milestone, the reconciliation document lists every design decision that the current model could not derive. We triage each such decision as a model gap (the calculus is extended), a model error (a claim is retracted, and the retraction is published), or an implementation artifact (no model change), and we report the count and disposition of these decisions, so that the model's revision history is visible.

Deliverables. Formal model and proofs; the link-class taxonomy as a citable specification; decision procedures in the form of a design questionnaire that maps an island's size, stake, threat model, and actor mix onto an

attainable bundle; and the *island design handbook* that consolidates them with the WP1 account and the WP3 translation table.

2.3 WP3 — Border checks: translating NFPs at island borders (M6–M48)

Method: protocol design with formal verification and conformance testing; in parallel with WP2.

Translation table. For every link class of Table 1 and every NFP-supporting concept (versions and hashes, identities and credentials, issuer lists, usage terms, encodings), WP3 determines what happens at each border direction: inbound from the open Web, outbound to it, and between two *islands*. Each entry is an upgrade, a downgrade, or a refusal. Upgrades happen at admission: a *Classic Jumplink* becomes a *Verified Link* once its target is snapshotted, hashed, and registered, and a claim becomes a *Credentialed Link* once an issuer that the island accepts has attested it. Downgrades happen on exit: a Verified or *Cryptographic Link* degrades to a Classic Jumplink for a non-member client, carrying a text fragment (Burris and Bokan 2023) that keeps the cited passage addressable where the page still contains it, and interoperates with Memento and Robust Links; a Cryptographic Link releases only its derived value; and a *Purpose-Bound Link* carries its terms, in ODRL with DPV purposes, so that they remain readable outside the island. Island F's border translates between binary protocol bindings and the uniform interface, following the WoT Binding Templates (Koster and Korkan 2025), so that its *Static Links* appear on the open Web as ordinary resources. Between two islands, translation maps one bundle onto another, e.g., whether Island L accepts a credential issued in Island M, based on the issuer lists and usage vocabularies that each island publishes as part of its NFP set. The governing rule is that every downgrade is visible: no reference may lose a guarantee silently.

Admission protocol. Inbound, an *admission protocol* (the *border crossing*) lets a client discover the island's NFP set and bind to it. The binding is a *credible commitment* in the sense of the W3C TAG's *User Agents* note (Yasskin and Capadisli 2026): the client's user agent restricts the use of the island's content, and nothing beyond it, and keeps that restriction after admission. Here the island applies its attribution regime. Clients present credentials in the W3C Verifiable Credentials 2.0 data model, with selective disclosure through SD-JWT (RFC 9901) or BBS-style signatures; browser clients request the presentation through the W3C Digital Credentials API (Caceres et al. 2026), so that the European Digital Identity Wallet can serve as a client from the start. The island publishes the issuers it accepts as part of its NFP set, and the registry checks revocation status at resolution time. Agents cross the same border through T4.3's gateway.

Formal conditions. In label terms, upgrades and downgrades are endorsement and declassification, and the border must satisfy a nonmalleability condition of the kind Cecchetti, Myers, and Arden (2017) formalize: no outsider may influence what the island releases, and the island may not endorse a claim whose content it cannot read. The second half of this condition forces WP2's model to treat admission under an anonymous credential explicitly, and it makes the border crossing a proof obligation. We state the visibility rule above as a property of the translation table and verify it for every entry.

Evaluation. A conformance test suite checks every table entry against the reference implementation, including adversarial inputs (replayed, revoked, or forged credentials; tampered snapshots; expired terms). We measure the latency and storage cost that each upgrade adds at admission, the share of outbound references that keep a resolvable passage after downgrade, measured on the WP5 corpora over the project period, and interoperability with the W3C specifications listed above.

Deliverables. Translation table v1 (M18) and v2 (M36); the admission protocol specification; border gateways for Islands L, S, M, and F; and the conformance test suite.

2.4 WP4 — Island technologies (M12–M54)

Method: systems research, cryptographic protocol design with proofs, and controlled comparative evaluation.
From the WP1 *islands* and the WP2 *link classes*, we deduce the technologies the islands require.

T4.1 Reference layer. We pin span-level identifiers to immutable versions in a Merkle-DAG store built on Git and IPFS primitives and on Software Heritage's ISO-standardized SWHIDs. Clients reach island resources through the Linked Web Storage protocol that the W3C LWS Working Group standardizes from Solid (Wright and Bremer 2026), and T4.1 contributes version-pinned identifiers to it as a profile. An island-scoped bidirectional link registry makes explicit the coordination that Xanadu wanted globally. Transclusion records provenance in a W3C PROV-compatible form. Every reference is typed by its WP2 link class, and typing is structural: resolving a Verified or *Cryptographic Link* returns the exact cited span, or a verified derivation of it, or fails visibly. Purpose-binding terms travel in the reference itself, expressed in the W3C ODRL policy language (Iannella and Villata 2018) with purposes from the Data Privacy Vocabulary of the W3C DPV Community Group (Golpayegani, Flake, and Ryan 2026), and the registry records issuance (to whom, for what, until when) as immutable history even though the derivation itself expires. We benchmark the implementation per link class on resolution latency, storage overhead, and link-maintenance cost, at island scales of 10^2 to 10^5

members. Baselines are a plain HTTP fetch for *Classic Jumplinks*, a Memento/Perma-style archival pipeline for *Archived Links*, and, for the registry, the measured editorial cost of manual citation archiving in Island L. Workloads come from the WP5 corpora.

T4.2 Confidential references. We generalize the PEP-LCA results (Grau 2025) into a *resolution protocol family*: references whose targets are secret-shared across island members and whose resolution is a joint computation with a verifiable result. SMPC has been deployed for single, fixed functions since the Danish sugar-beet auction (Bogetoft et al. 2009) and in Estonian government statistics (Bogdanov et al. 2016), and general-purpose protocols and implementations such as SPDZ (Damgård et al. 2012) and MP-SPDZ (Keller 2020) make arbitrary circuits practical. We build on MP-SPDZ and contribute what these systems lack: a resolution protocol bound to a citable, version-pinned reference. The nearest system precedent, Fabric (§2.2), places cleartext only on nodes the data owner trusts; our protocol family removes that requirement, since no node holds a Cryptographic Link's target in full. Viaduct (Acay et al. 2021) shows that the choice among multi-party computation, zero-knowledge proofs, and commitments can be derived from labels automatically; we take the same stance at the granularity of a reference, with the WP2 link class selecting the protocol. Three arcs run partly in parallel. *(i) From honest-but-curious to malicious security:* we use verifiable secret sharing and zero-knowledge correctness proofs, and state and prove security in standard simulation-based models against a static, then an adaptive, adversary; the deliverable is a quantified privacy-performance frontier. *(ii) Breaking availability coupling:* in the PEP-LCA deployment, resolution requires enough members to be online simultaneously, which obstructs interactive use; we investigate threshold reconstruction and precomputed correlated randomness so that resolution tolerates member downtime, and we treat the resulting availability guarantee as an explicit WP2 coordination requirement. *(iii) Input integrity:* committed inputs and attested sources that connect to C2PA-style hardware roots make a verified derivation over garbage inputs distinguishable from one over attested inputs. We measure performance on topologies drawn from Island S's actual supply-chain structure and report resolution latency against member count, share-reconstruction threshold, and circuit depth. K. Mitrokotsa and N. Cheng, who co-supervised the PEP-LCA work, continue as external cryptographic collaborators.

T4.3 Agents and static bindings. The island's agent gateway exposes capabilities as state-dependent *affordances*, through a HATEOAS-style layer over MCP-class agent protocols, with semantics for *why* an action is currently valid and explicit ordering guarantees for dependent state-changing calls. Each affordance carries the WP2 link class of the resource it exposes, so that an agent can distinguish a Verified citation from a Classic Jumplink before acting. An agent crosses the WP3 border by accepting the island's NFP set and presenting a *verifiable credential* that states on whose behalf it acts and within which limits, so that identity, delegated authority, and obligations bind before any state-changing call. For Island F, the same descriptions bind interfaces at design time as *Static Links* over binary protocol bindings. *Evaluation design:* we compare, within each scenario, (a) statically linked tool catalogues, (b) our affordance layer, (c) hybrids, and (d) Static Links. Scenarios come from Islands L, S, and M and from Island F's installations in building automation (Siemens) and edge computing (AFarEdge). Dependent measures are task success rate; latency per interaction; safety violations, operationalized as state-changing actions on references whose declared class does not support them, ordering violations on dependent calls, and the attack success rate under the indirect-prompt-injection tasks of AgentDojo (Debenedetti et al. 2024; Greshake et al. 2023), adapted to island scenarios; and evolvability, measured by re-running a fixed task suite after a scripted change to the island's capability surface and recording the share of tasks that succeed without client modification. Model choice, prompt, and random seed are held fixed across conditions and reported. **Hypotheses:** affordance-scoped exposure outperforms statically linked catalogues on safety and evolvability without a significant loss in task success; Static Links outperform both on latency in installations that do not change. Together, the conditions measure the trade-off that the WP2 link-time axis captures, and hence which *link time* the calculus should recommend for a given installation. We treat a negative result as publishable (see §5). Standardization proceeds through the W3C Autonomous Agents on the Web Community Group, extending our work on *signifiers*, and through the AI Agent Protocol Community Group for the agent credential. Static Links go to the W3C Web of Things Working Group as a profile of its protocol binding templates, through the Web of Things channel that the PI's Chips JU role already funds (§3).

2.5 WP5 — Validation islands and incentives (M12–M60)

Method: three case studies, an industrial demonstrator, and a repeated stakeholder analysis; runs in parallel with WP1 from M12.

Community involvement from M12. WP5 starts while WP1 is still running. At M12, when WP1 delivers its first account of the required *islands*, we convene one community panel per island: the editorial partner, legal practitioners, and Open Terms Archive for Island L, the WISER partners for Island S, public-interest publishers and instance operators for Island M, nanopublication server operators and research infrastructures for Island

N, and the AFarEdge and Siemens installation owners for Island F. Each panel reviews the NFP bundle and the exclusions of its island, names the actors who would keep each guarantee, and agrees on the corpus and the measures used later in the island. Panel feedback enters the WP1 revision at M18, so that the communities shape the islands before WP2–WP4 fix the technologies. The panels meet again at each integration milestone and supply the interviewees for the incentive re-assessment.

Island L (legal citation and copyright). In an end-to-end pilot, the citations in a working corpus become span-level, version-pinned, and verifiable. The corpus is a real editorial pipeline (a law journal's, or a moot-court record) **[TODO: confirm the partner corpus and its data-sharing terms]**. *Verified Links* anchor to ECLI and ELI identifiers where the cited source has one, so the *island* adds span and version pinning to the EU's document-level identifiers. We measure the citation-integrity failures made structurally impossible, against a baseline audit of the same corpus; the editorial cost per citation, compared to Perma-style manual archiving; and practitioner acceptance, through structured interviews and task observation with the editorial team. Island L also instantiates the *attribution* NFP and weighs verified authorship against pseudonymous regimes. *Credentialed Links* let an author prove membership of the bar, or a court the origin of a judgment, without the island holding a full identity record; the partner group's doctoral work on chilling effects grounds the anonymity side. Our partner is Prof. Tamò-Larrieux (Lausanne), and Island L docks onto the PI's running SNSF CoCoDa and *Access to Justice* projects. Matti Schneider (Open Terms Archive), whose archive records the terms and DSA reports of digital services and which the European Commission's DSA enforcement team uses, is a potential partner: Island L would let providers publish these documents as signed, versioned records that the archive no longer has to reconstruct **[TODO: confirm with M. Schneider]**.

Island S (supply chain). We extend the running GDT/PEP-LCA deployment into a full island with confidential references and agent-mediated onboarding. Empa, where Jan Grau conducts his PhD under the PI's supervision, contributes the life cycle assessment (LCA) methods, i.e., which emission and footprint figures the island computes over its members' confidential data and how verifiers check them, and Siemens serves as industrial validation site. Its workloads follow the data duties of the EU battery passport (Regulation (EU) 2023/1542, Art. 77, from February 2027) and of CBAM (Regulation (EU) 2023/956), which both require emissions data from upstream suppliers. Under CBAM, actual emission values must be checked by accredited verifiers (Implementing Regulation (EU) 2025/2546; Delegated Regulation (EU) 2025/2551), so Island S also tests Credentialed Links that carry the verifier's attestation alongside a confidential value. Because the deployment already exists, Island S provides early evidence: it exercises T4.2 protocols on real topologies well before Island L's editorial integration is complete.

Island M (platforms). Public-interest publishers (e.g., public broadcasters, election authorities, and newsrooms) form an island on an ActivityPub-based federated network. Their posts become version-pinned and attributed, edits and deletions become superseding records, and Credentialed Links state who stands behind an account. *Purpose-Bound Links* carry the terms under which model providers may use the posts. Vetted researchers resolve *Cryptographic Links* to exposure statistics that instance operators compute jointly over interaction data that none of them discloses, which tests T4.2 against the researcher data access of DSA Art. 40. We measure the survival of posts and links compared to platform baselines, including the reachability of public platform content for logged-out clients over the project period; the share of research queries answered without raw-data transfer; and acceptance by publishers and researchers. **[TODO: name the publisher and instance-operator partners]**

Island F (fixed installations). In building automation (Siemens) and edge computing (AFarEdge), we deploy *Static Links* over binary protocol bindings behind a WP3 border gateway and measure latency, the share of interactions that bypass run-time discovery, and the cost of a controlled change to an installation, which a Static Link must surface as a visible failure.

Incentive re-assessment and business, legal, and economic evaluation. In the last project year, we repeat the WP1 interviews with each class of actor and measure whether the deployed islands changed their incentive to centralize, and whether the guarantee-respecting path became their least costly one. We also evaluate each deployed island's artifacts against the WP1 business, legal, and economic analyses: whether its operating costs are covered by the business model foreseen, whether its records held up against the legal obligations of its domain, and how its costs and benefits fell on each actor.

3. Feasibility

Most of the work packages extend systems that are already running.

Preliminary results.

- *Island drafts and stakeholder access (WP1).* The island drafts rest on running work: the legal-domain projects behind Island L, the GDT/PEP-LCA deployment behind Island S, our platform-transparency and

DSA Article 40(12) audit work behind Island M, the PI's Web of Things architecture work in AFarEdge behind Island F, and the deployed nanopublication network behind Island N. Our published audit instruments supply a validated method for the stakeholder analysis.

- *Confidential resolution (T4.2).* The PEP-LCA protocol computes joint supply-chain assessments across up to 50,000 participants with no trusted third party, which shows that the scale T4.2 targets is reachable. Production SMPC deployments elsewhere (Bogetoft et al. 2009; Bogdanov et al. 2016) show that the approach works with real institutions. The open problems are the security model, availability coupling, and input integrity, which T4.2's arcs address.
- *SMPC over pods (T4.1/T4.2, WP3).* Our Solid Symposium 2026 system anchors SMPC share disclosure in an access-control layer with WASM-sandboxed data transformation, demonstrating the confidentiality-boundary integration that the island architecture and its border checks assume.
- *Agent affordances (T4.3).* Our work on *signifiers* (AAMAS 2023/2024) has delivered first-class run-time action-repertoire abstractions and BDI agents that reason over them, so T4.3 begins from working systems.
- *Standardization (WP3/WP4).* The PI's contributions to the W3C Web of Things Thing Description are deployed in shipped industrial products. The PI's Chips JU role also funds and schedules a W3C community report on Web of Things application, due within the WP3 and WP4 windows.

Institutional feasibility. Running projects dock onto the work and secure site access: Innosuisse Flagship WISER (supply-chain LCA, Island S), SNSF CoCoDa (VLOP regulation, with the ODI as partner, for the Solid and governance line), and SNSF/Innosuisse *Access to Justice* (the legal-domain relationship behind Island L). The Chips JU action **AFarEdge** (2026–2029) docks onto T4.3 and Island F, which would otherwise be the least institutionally anchored part of the programme: the PI leads the definition of its Web-of-Things architecture for heterogeneous edge hardware and cognitively heterogeneous agents, which gives the agent layer and the downgrade island an industrial reference point, a standardization channel, and a working relationship with silicon and platform vendors. As General Chair of the Solid Symposium 2027 and of the ACM Symposium on Computer Science and Law 2028, the PI convenes two of the communities WP1 interviews. The PI stepped down as Dean in February 2026 to refocus on research, which frees the time that the ≥40% commitment requires. As a business school, HSG also hosts the business and economic expertise the WP1 and WP5 analyses need, including Tim Meyer's professorship in decentralized business models, and FuSo provides the long-standing collaboration with legal scholars.

Team feasibility. The two researchers who built the systems the project draws on join from M1. Luka Bekavac conducts WP1's stakeholder interviews, adapts the platform-audit instruments he developed to Island M's research-access measurements, and convenes the community panels of Islands L and M at M12; he completes his PhD at about M13 (May 2029) and stays as a postdoc until M19, with Postdoc B overlapping from M17. Jan Grau, who is employed by Empa and conducts his PhD there under the PI's supervision until about May 2030 (≈M25), contributes in kind: he extends his PEP-LCA protocol into the first, honest-but-curious arc of T4.2 and runs it on Island S's supply-chain topologies with Empa and Siemens; Postdoc A works with him from M1 and leads T4.2 after he completes his PhD (Resources and Time Commitment, §2). ISLANDS is not a collaborative project: the PI's team leads and conducts every work package, and we name an unfunded collaborator only where a minor part needs expertise outside the PI's record, i.e., Mitrokotsa and Cheng for the malicious-security proofs of T4.2, Tamò-Larrieux for the legal analysis and the co-supervision on Island L, Tim Meyer for the business models of the islands, an economist at HSG for the economic analysis, and Empa for the life cycle assessment (LCA) methods behind Island S. -->

4. Work plan

4.1 Deliverables and milestones

ID	Deliverable	WP	Due
D1.1	Account of societally required islands v1 + architectural decision record v1	WP1	M12
D1.2	Stakeholder analysis report; account and decision record v2	WP1	M18
D2.1	Guarantee calculus v0 + link-class taxonomy spec	WP2	M18
D2.2	Calculus v1, revised against first measurements	WP2	M36
D2.3	Island design handbook	WP2	M60
D3.1	Translation table v1 + admission protocol spec	WP3	M18

ID	Deliverable	WP	Due
D3.2	Border gateways for Islands L, S, M, F + conformance suite	WP3	M36
D3.3	Translation table v2, verified visibility property	WP3	M48
D4.1	Reference layer prototype (T4.1) + agent gateway v1 (T4.3)	WP4	M24
D4.2	Resolution protocol family, honest-but-curious (T4.2)	WP4	M24
D4.3	Malicious-secure arc + privacy-performance frontier (T4.2); comparative agent evaluation (T4.3)	WP4	M42
D4.4	Availability-decoupled resolution + input integrity (T4.2); specification to W3C Autonomous Agents on the Web CG (T4.3)	WP4	M54
D5.0	Community panels convened per island; corpora and measures agreed	WP5	M15
D5.1	Islands L, S, M instantiated; Island F demonstrator	WP5	M30
D5.2	First measurement campaign	WP5	M42
D5.3	Final evaluation + incentive re-assessment	WP5	M60
M-I1..I5	Annual integration milestone: calculus reconciliation, revised WP1 account and decision record	all	M12/24/36/48/60

4.2 Calendar

Indicative start 1 May 2028 (M1).

- **Year 1** (M1–M12): *island* drafts solidified, stakeholder interviews, account and decision record v1 (WP1); calculus and translation work begin (WP2, WP3, from M6); hires 1–4 in post; islands scoped with partners. **I1.**
- **Year 2** (M13–M24): community panels per island convened and feeding the WP1 revision (WP5); stakeholder report (WP1); calculus v0 and translation table v1 (WP2, WP3); reference layer, honest-but-curious resolution, and agent gateway v1 (WP4); Postdoc B in post from ~M17, Bekavac's handover completed with overlap by M19 (November 2029) and Grau's by M25 (completion of his PhD, May 2030); PhD 3 in post from ~M21. **I2.**
- **Year 3** (M25–M36): border gateways and conformance suite (WP3); malicious-secure arc (T4.2); comparative agent evaluation begins (T4.3); islands instantiated (WP5); calculus v1 revised against evidence. **I3.**
- **Year 4** (M37–M48): translation table v2 (WP3); availability decoupling (T4.2); cross-island interoperation; first measurement campaign (WP5); specifications to W3C. **I4.**
- **Year 5** (M49–M60): final evaluation and incentive re-assessment (WP5); island design handbook (WP2); standardization handoff, open artifacts. **I5.**

4.3 Critical path and dependencies

WP1 is the entry point, but it does not block the others: the *island* drafts are concrete enough for WP2 and WP3 to start at M6, and T4.2 continues the running PEP-LCA work from the start. The critical path runs from T4.2(i) through T4.2(ii) to D4.4, because availability-decoupled malicious-secure resolution is both the hardest result and the one on which Island S's sustained operation depends. WP3 depends on WP2's taxonomy (D2.1) but not on its proofs, so that the two run in parallel and reconcile at each integration milestone. T4.1 can be scheduled independently of T4.2 until the *Cryptographic Link* class is integrated at I2. T4.3 depends on T4.1's link typing but not on T4.2's completion, since it can expose Cryptographic Links as opaque *affordances* before the underlying protocols are malicious-secure. Island S can begin measurement before T4.1 is complete; Island L cannot, since its value lies in span-level verifiable citation.

5. Risk assessment, mitigation, and contingency

For each risk, we state its likelihood (L) and impact (I), the *mitigation* that lowers the likelihood, and the *contingency* that we apply if the risk materializes anyway. Each contingency keeps a publishable result in reach, and the annual integration milestones are the points at which we decide whether to switch to it.

#	Risk	L	I	Mitigation (before)	Contingency (if it occurs)
R1	Coordination cost exceeds what islands of realistic size can sustain, so the top link classes are unreachable in practice	M	H	T4.1 instruments coordination cost from the first prototype, so the problem shows at I1; WP2's calculus is stated over bundles, not only the top rung	Concentrate on the attainable rungs (*Trusted*, *Archived*, *Verified Links*) and publish the validated boundary of attainable bundles; WP2 returns "not attainable" as a first-class result
R2	SMPC resolution latency stays too high for interactive use	M	H	Preprocessing and precomputed correlated randomness in the SPDZ/MP-SPDZ model (T4.2 ii); latency measured on Island S topologies from M24	Resolve *Cryptographic Links* asynchronously and cache the verified derivations; keep interactive resolution for small islands and run Island S in batch mode; the privacy-performance frontier (D4.3) stands either way
R3	The malicious-secure arc does not close within budget	M	M	Staged security goals (static, then adaptive adversary) with publishable intermediate states	Deliver honest-but-curious resolution with covert security and zero-knowledge proofs for selected circuits; D4.2 already exceeds the deployed state of the art
R4	**Purpose binding is not technically enforceable after resolution**	H	M	Designed as an economic guarantee: WP2 scores it by the margin by which re-requesting is less costly than hoarding	Drop *Purpose-Bound Links* from islands whose actors show a negative compliance margin and record this in the decision record; the class remains in the calculus as a priced guarantee
R5	Loss of a validation site (editorial partner withdraws; industrial partner reorganizes)	M	H	Five islands with separate partners; community panels from M12 bind partners early; Island S sits in a running deployment	Island L moves to an alternative corpus (law journal, moot-court record, *Access to Justice*) **[TODO: secure a written second option]**; Island S moves to another WISER partner; Island F to the AFarEdge consortium; Island M to other public-interest publishers
R6	The agent-layer comparison returns a negative or null result	M	L	Pre-registered hypotheses and analysis plan	Publish the null result; the link-time measurements (static vs. run-time binding) still inform WP2's link-time axis
R7	The applied-cryptography postdoc cannot be recruited (hardest profile)	M	H	International search from the Step 2 result (scheduled for 17 December 2027), about four months before M1, at the IACR job board, CRYPTO, Eurocrypt, and ACM CCS and through the networks of Mitrokotsa and Cheng	Convert the position into two PhD positions with more advisory time from Mitrokotsa and Cheng; Grau's SMPC expertise covers T4.2's first arc in the meantime
R8	Standardization uptake does not materialize	M	M	W3C groups named per work package (§1); the PI's specifications have reached shipped products before (WoT TDs)	Publish the specifications as W3C Community Group reports and open specifications with the reference implementation, which stand independently of adoption

#	Risk	L	I	Mitigation (before)	Contingency (if it occurs)
R9	Legal exposure from the Island L corpus (personal data in citations)	L	M	Retraction as a superseding record (GDPR Art. 17); DPIA at island scoping; legal partner embedded **[TODO: confirm USG ethics/DPIA route]**	Restrict the corpus to published judgments and legislation, and pseudonymize parties before ingestion
R10	WP1 finds that society requires islands beyond the five drafts, or that a draft is not required	M	M	Account and decision record revised annually; WP2's calculus is stated over NFP bundles, not over the drafts	Replace a validation island through the decision record and redirect its WP5 effort; the dropped draft and its reason are themselves a result
R11	Some NFPs cannot be translated at a border without loss	H	L	The visibility rule of WP3 forbids silent loss	Record such entries as refusals or visible downgrades; cross-island traffic for these classes falls back to *Classic Jumplinks*, and the table of untranslatable entries bounds what islands promise across borders
R12	Providers and platforms decline to publish into an island (e.g., terms of service for Island L, posts for Island M)	H	M	Islands start with communities that have an incentive to join (public-interest publishers, research infrastructures, supply-chain partners under disclosure duties); WP1 analyses each actor's incentives	A trusted third party, such as Open Terms Archive for terms of service, joins the island and publishes signed snapshots of what providers publish elsewhere; references then hold *Archived* or *Trusted* instead of *Verified* guarantees, which the calculus expresses
R13	Community panels do not engage from M12	L	M	Panels build on existing relationships (WISER, *Access to Justice*, CoCoDa, AFarEdge, Open Terms Archive) and on the Solid Symposium 2027 and CS&Law 2028, which the PI chairs	Proceed with partner-only panels and recruit further members through the PI's conferences and the W3C groups; WP1 records the gap in the account
R14	Reviewers or communities read Island L as digital rights management	M	M	Island L records reuse in published works, not who reads, and restricts neither reading nor copying; the decision record excludes islands that enforce usage on members' devices; admission follows the W3C TAG's loyalty duty of user agents (Yasskin and Capadisli 2026)	If a community asks for enforcement, the request is recorded as an exclusion and Island L proceeds with attribution and verifiable terms only

In R1, R4, R6, R11, and R12, the likely failure is a bounded negative result, and each contingency delivers that result as a finding, as fits a project whose core contribution is a theory of what is attainable.

References (not counted towards the page limit)

Acay, Recto, Gancher, Myers, Shi. Viaduct: An extensible, optimizing compiler for secure distributed programs. *PLDI 2021*, 740–755. doi:10.1145/3453483.3454074

Burris, Bokan (eds.). *URL Fragment Text Directives.* W3C WICG Draft Community Group Report, 2023.

Caceres, Cappalli, Yosef, Goto (eds.). *Digital Credentials.* W3C Working Draft, Federated Identity Working Group, 2026.

Golpayegani, Flake, Ryan (eds.). *Data Privacy Vocabulary (DPV)*, version 2.3. W3C DPV Community Group Final Report, 25 February 2026.

Iannella, Villata (eds.). *ODRL Information Model 2.2.* W3C Recommendation, 15 February 2018.

Koster, Korkan (eds.). *Web of Things (WoT) Binding Templates.* W3C Group Note, 4 November 2025.

Wright, Bremer (eds.). *Linked Web Storage Protocol 1.0.* W3C Working Draft, Linked Web Storage Working Group, 2026.

Reddit, Inc. Q3 2026 earnings call, 29 October 2026 (transcript: The Motley Fool, 30 October 2026); TechCrunch, 29 October 2026.

Capadisli, Odelola, Guy, Nitot (eds.). *Self-Review Questionnaire: Societal Impact.* W3C TAG Group Note, 28 October 2026.

Yasskin, Capadisli (eds.). *User Agents.* W3C TAG Group Note, 22 October 2026.

Bogdanov, Kamm, Kubo, Rebane, Sokk, Talviste. Students and taxes: A privacy-preserving study using secure computation. *PoPETs* 2016(3):117–135. doi:10.1515/popets-2016-0019

Bogetoft, Christensen, Damgård, Geisler, Jakobsen, Krøigaard, et al. Secure multiparty computation goes live. *FC 2009*, LNCS 5628:325–343. doi:10.1007/978-3-642-03549-4_20

Cecchetti, Myers, Arden. Nonmalleable information flow control. *ACM CCS 2017*, 1875–1891. doi:10.1145/3133956.3134054

Damgård, Pastro, Smart, Zakarias. Multiparty computation from somewhat homomorphic encryption. *CRYPTO 2012*, LNCS 7417:643–662. doi:10.1007/978-3-642-32009-5_38

Debenedetti, Zhang, Balunović, Beurer-Kellner, Fischer, Tramèr. AgentDojo: A dynamic environment to evaluate prompt injection attacks and defenses for LLM agents. *NeurIPS 2024* (Datasets and Benchmarks), 82895–82920.

Grau. Towards decentralized privacy-enhancing life cycle assessment. MSc thesis, University of St. Gallen, 2025.

Greshake, Abdelnabi, Mishra, Endres, Holz, Fritz. Not what you've signed up for: Compromising real-world LLM-integrated applications with indirect prompt injection. *ACM AISec '23*, 79–90. doi:10.1145/3605764.3623985

Keller. MP-SPDZ: A versatile framework for multi-party computation. *ACM CCS 2020*, 1575–1590. doi:10.1145/3372297.3417872

Fett, Yasuda, Campbell. *Selective Disclosure for JSON Web Tokens.* RFC 9901, IETF, 2025.

Kuhn, Dumontier. Trusty URIs: Verifiable, immutable, and permanent digital artifacts for linked data. *ESWC 2014*, LNCS 8465:395–410.

Kuhn, Chichester, Krauthammer, Queralt-Rosinach, Verborgh, et al. Decentralized provenance-aware publishing with nanopublications. *PeerJ Computer Science* 2:e78, 2016.

Nanopublication Association. *Nanopublications.* nanopub.net, accessed 5 November 2026.

Levine. *Linkers and Loaders.* Morgan Kaufmann, 2000.

Sporny, Thibodeau, Herman, Cohen, Jones (eds.). *Verifiable Credentials Data Model v2.0.* W3C Recommendation, 15 May 2025.

Liu, George, Vikram, Qi, Waye, Myers. Fabric: A platform for secure distributed computation and storage. *SOSP 2009*, 321–334. doi:10.1145/1629575.1629606

Myers, Liskov. A decentralized model for information flow control. *SOSP 1997*, 129–142.

European regulation cited by number in the text: Regulation (EU) 2016/679 (GDPR), Art. 17; Regulation (EU) 2023/1542 (Batteries), Art. 77; Regulation (EU) 2023/956 (CBAM); Regulation (EU) 2022/2065 (DSA); European Commission, decision fining X under the DSA, 5 December 2025; Council conclusions on ECLI (OJ C 127, 2011) and ELI (OJ C 325, 2012).

Annex — Ongoing and submitted grants

[TODO] Complete per the current template: ongoing and submitted grants with role, funder, reference number, period, amount, and percentage of PI time, plus an explicit statement of non-overlap with ISLANDS. Entries to include: Innosuisse Flagship WISER (2022–2026); SNSF CoCoDa (2025–2029); SNSF/Innosuisse *Access to Justice* (2026–2030); EU Chips JU AFarEdge (2026–2029). For AFarEdge, state the effort explicitly (39.5 person-months over 36 months, ≈ 0.09 FTE) so that it reads against the ≥40% ISLANDS commitment as negligible rather than as competing load, and resolve the funding status before stating any amount (see TODO 4). For each, state which parts of the ISLANDS work programme are adjacent and why they are not redundant, since WISER and CoCoDa both dock onto WP5 and a reviewer will look for double funding.